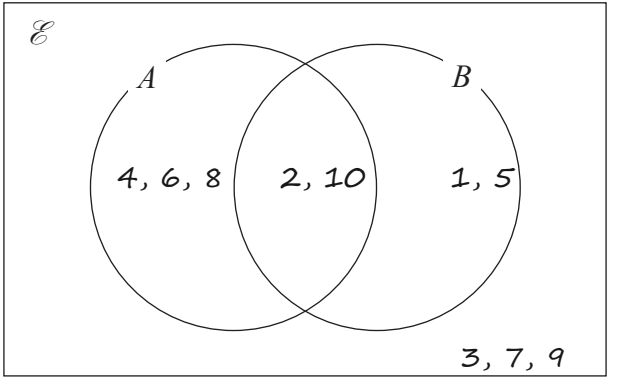
22 $\mathscr{E} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$
$A = \{\text{even numbers}\}$
$B = \{\text{factors of } 10\}$

(a) Complete the Venn diagram for this information.



(3)

3 marks total
A only: 4, 6, 8
Both: 2, 10
B only: 1, 5; outside: 3, 7, 9

A number is chosen at random from the universal set, $\mathscr{E}$

(b) Find the probability that this number is in the set $A \cap B$

A ∩ B contains 2 and 10: two outcomes.

Probability = 2/10 = 1/5.

2 marks total
2 favourable of 10
2/10 or 1/5

(2)

(Total for Question 22 is 5 marks)

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS A